CSP-203: Software Tools & Technologies

Laboratory

Learning Objectives:-
1.Proficiency in Document Structure
2.Mathematical Environments
3.Custom Commands & Macros
4.Using Packages
5.Tables, Lists, and Sections
6.Problem Solving with LaTeX

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Advanced LaTeX Topics

1. Title, Author, Abstract

Input:

```
\documentclass{article}
\title{The Title}
\author{A. Author}
\date{\today}
\begin{document}
\maketitle
\begin{abstract}
Abstract goes here...
\end{abstract}
\end{document}
```

The Title

A. Author

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Abstract

Abstract goes here...

Explanations: - `\title` → Defines the document's title, stored in memory until printed.
- `\author` → Specifies the author's name(s) for the title block.
- `\date` → Sets the date (use `\today` for automatic current date).
- `\maketitle` → Generates the formatted title page using title/author/date.
- `abstract` → Creates a dedicated block for a short paper summary.

2. Sections and Subsections

Input:

```
\documentclass{article}
\begin{document}
\section{Introduction} The problem of \ldots
\section{Method}
\subsection{Sample Preparation}
\subsection{Data Collection}
\section{Results}
\section{Conclusion}
\end{document}
```

1 Introduction

The problem of …

2 Method

2.1 Sample Preparation

2.2 Data Collection

3 Results

4 Conclusion

Explanations: - `\section` → Creates a main section heading with automatic numbering.
- `\subsection` → Creates a subsection heading under the current section.
- `\subsubsection` → Creates a deeper level heading under a subsection.
- `\paragraph` → Creates a smaller heading inline with paragraph text.
- `\subparagraph` → Creates the smallest heading, often inline with text.
- Adding * (e.g., `\section*`) → Creates the same heading but without numbering or inclusion in the Table of Contents.

3. Cross-References

Input:

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
\section{Introduction}
\label{sec:intro}
In Section \ref{sec:method}, we \ldots
\section{Method}
\label{sec:method}
\begin{equation}
\label{eq:euler} e^{i\pi} + 1 = 0
\end{equation}
By \eqref{eq:euler}, we have \ldots
\end{document}
```

5 Introduction

In Section 6, we …

6 Method

$$e^{i\pi} + 1 = 0 \tag{1}$$

By (1), we have …

Explanations: - `\label` → Assigns a unique tag to a section, figure, table, or equation for later referencing.
- `\ref` → Inserts the automatically generated number of a tagged element (e.g., section, figure, table).
- `\eqref` → Specifically references equations and automatically adds parentheses around the number.

4. Figures with Labels

Input:

```
\documentclass{article}
\usepackage{graphicx}
\begin{document}
Figure \ref{fig:sample} shows \ldots
\begin{figure}
\centering
\includegraphics[scale=0.3]{example-image}
\caption{\label{fig:sample}Sample figure}
\end{figure}
\end{document}
```

Output:

Figure 1 shows . . .



Figure 1: Sample figure

`\ref{fig:sample}` → Retrieves and prints the number assigned to the labeled figure (here "fig:sample"), allowing automatic cross-referencing.
- `\caption` + `\label` → Adds a numbered caption to the figure and assigns a reference label for cross-referencing.

Explanations: - `figure` → A floating environment used to include and position images in the document.
- `\includegraphics` → Command from the graphicx package that loads and displays an external image file.

5. Tables

Input:

```
\begin{tabular}{|l|c|r|} \hline
Item    & Quantity & Unit Price (\$) \\ \hline
Widget  & 1        & 199.99 \\ \hline
Gadget  & 2        & 399.99 \\ \hline
Cable   & 3        & 19.99  \\ \hline
\end{tabular}
```

Output:

Item	Quantity	Unit Price ($)
Widget	1	199.99
Gadget	2	399.99
Cable	3	19.99

Explanations:

- `tabular{...}` → environment for creating tables.

- `l c r` → column alignment: left, center, and right respectively.

- Vertical bars `|` → draw borders between columns.

- First row → header row (Item, Quantity, Unit Price).

- `\hline` → inserts a horizontal rule above or below a row.

- `&` → separates columns inside a row.

- \\ → ends the current row and moves to the next.

Algorithms in LaTeX

Input

```
\documentclass{article}
\usepackage{algorithm2e}
\begin{document}
\begin{algorithm}
$i \gets 10$\;
\eIf{$i \geq 5$}
 { $i \gets i-1$\; }
 { \If{$i \leq 3$}
   { $i \gets i+2$\; }
 }
\end{algorithm}
\end{document}
```

Output

$i \leftarrow 10;$
if $i \geq 5$ **then**
  $i \leftarrow i - 1;$
else
  if $i \leq 3$ **then**
    $i \leftarrow i + 2;$
  end
end

Explanations

- `\usepackage{algorithm2e}` → loads the package for typesetting pseudocode.
- `\begin{algorithm}...\end{algorithm}` → creates an algorithm block.
- `$i \gets 10$` → assigns 10 to variable i.
- `\;` → ends the current statement and moves to the next line.
- `\eIf{condition}{true block}{false block}` → defines an if–else structure.
- `\If{condition}{block}` → defines a single if condition without an else.

Bibliography with BibTeX (TeX Live Method)

Step 1: Create a `.bib` file

In the same folder as your `.tex` file, create `bib-datasci.bib`:

```
@book{Brooks1997Methodology,
  author    = {Brooks, Frederick P.},
  title     = {The Mythical Man-Month: Essays on Software
     Engineering},
  year      = {1997},
  publisher = {Addison-Wesley}
}

@article{Jacobson1999Towards,
  author  = {Jacobson, Ivan},
  title   = {Towards Unified Modeling},
  journal = {Journal of Object-Oriented Development},
  year    = {1999},
  volume  = {12},
  number  = {3},
  pages   = {45--50}
}
```

Step 2: Write your main `.tex` file

Add the citations and bibliography commands:

```
\documentclass{article}
\usepackage{natbib}

\begin{document}

As shown in \citet{Brooks1997Methodology}, software projects
often face challenges \citep{Jacobson1999Towards}.

\bibliographystyle{plainnat}
\bibliography{bib-datasci}

\end{document}
```

Step 3: Compile using TeX Live + TeXstudio

BibTeX requires multiple runs:

1. Press **F5** (Build & View). You will see [?] instead of citations.
2. Press **F8** (Tools → Bibliography) to run BibTeX.
3. Press **F5** two more times.

Result: Citations now appear correctly, e.g.,

As shown by Brooks et al. (1997), software projects often face challenges (Jacobson, 1999).

Step 4: Key Commands Recap

- `\usepackage{natbib}` → enables flexible citations.
- `\citet{key}` → Author (year) style.
- `\citep{key}` → (Author, year) style.
- `\bibliographystyle{plainnat}` → sets citation format.
- `\bibliography{file}` → loads references from `file.bib`.

Custom Macro Example – Display Math Version

Input:

```
\documentclass[12pt]{IEEEtran}
\usepackage{amsmath}

% Define a macro for the performance ratio
\newcommand{\rperf}{\rho_{\text{perf}}}

\begin{document}

The performance ratio is displayed as
\[
\rperf = {\bf c}'{\bf X} + \varepsilon
\]

\end{document}
```

Output:

The performance ratio is displayed as

$$\rho_{\text{perf}} = \mathbf{c}'\mathbf{X} + \varepsilon$$

Explanations:

- `\newcommand{\rperf}{\rho_{\text{perf}}}` → Defines a shortcut macro " that expands to ρ_{perf}.
- `\rperf` → Calls the macro to produce ρ_{perf} wherever it is used.
- `\[ ... \]` → Puts the equation in display math mode (centered on its own line).
- Custom macros reduce repetition, prevent typos, and maintain consistent notation.